CI/CD Demo

File Structure:

```
cicd-demo/
├── .github/
│   └── workflows/
│       └── pipeline.yml
├── site/
│   └── index.html
├── app.py
└── test_app.py
```

app.py

```
def add(a, b):
    return a + b

def is_even(n):
    return n % 2 == 0

if __name__ == "__main__":
    html = f"<h1>CI/CD Demo</h1><p>add(2, 3) = {add(2, 3)}</p><p>is_even(4) = {is_even(4)}</p>"
    open("site/index.html", “w").write(html)
```

test_app.py

```
from app import add, is_even

def test_add():
    assert add(2, 3) == 5

def test_is_even():
    assert is_even(4)
    assert not is_even(7)
```

site/index.html

```
<!DOCTYPE html>
<html><body>
  <h1>CI/CD Demo</h1>
</body></html>
```

pipeline.yml

```
name: CI/CD

on: push

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: "3.12"
      - run: pip install pytest
      - run: pytest

  deploy:
    needs: test
    runs-on: ubuntu-latest
    permissions:
      pages: write
      id-token: write
    environment: github-pages
    steps:
      - uses: actions/checkout@v4
      - run: python3 app.py
      - uses: actions/upload-pages-artifact@v3
        with:
          path: site
      - uses: actions/deploy-pages@v4
```

GitHub Setup

Create a new public Repository
Go to Settings → Pages → Build and deployment → Source and choose GitHub Actions.
After the first deploy, check Settings → Environments → github-pages (optional). Untick Required reviewers if you want fully automatic deploys.

Push from Terminal

git init
git add .
git commit -m "initial commit"
git branch -M main
git remote add origin GitHub Repo link
git push -u origin main

Check Site
Settings → Pages: "Your site is live at: " with a Visit site button.

Change to a-b in app.py and push to see the CI/CD pipeline fail as it does not pass the test.
Change it back to a+b to see it deploy again